sai_pavani@Sai-Pavani: ~/OS_

GNU nano 8.7.1 skill7.c *

```c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    char *path = getenv("PATH");

    if (path == NULL) {
        printf("PATH variable not found.\n");
        return 1;
    }

    printf("PATH Variable:\n%s\n\n", path);

    pid_t pid = fork();

    if (pid < 0) {
        perror("fork failed");
        return 1;
    }

    if (pid == 0) {
        // Child process
        printf("Child Process Started\n");
        printf("Child PID: %d\n", getpid());

        // Execute ls
        execlp("ls", "ls", NULL);

        // Runs only if exec fails
        perror("exec failed");
        exit(1);
    } else {
        // Parent process
        int status;

        printf("Parent waiting for child...\n");
        waitpid(pid, &status, 0);

        if (WIFEXITED(status)) {
            printf("Child exited with status: %d\n", WEXITSTATUS(status));
        }
    }

    return 0;
}
```

```
sai_pavani@Sai-Pavani:~/OS_SKILL$ nano skill7.c
sai_pavani@Sai-Pavani:~/OS_SKILL$ gcc skill7.c -o skill7
sai_pavani@Sai-Pavani:~/OS_SKILL$ ./skill7
PATH Variable:
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/WindowsApps/MicrosoftCorporationII.WindowsSubsystemForLinux_2.7.11.0_x64__8wekyb3d8bbwe:/mnt/c/Program Files/Java/jdk-25/:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Windows/system32:/mnt/c/Windows:/mnt/c/Windows/System32/Wbem:/mnt/c/Windows/System32/WindowsPowerShell/v1.0/:/mnt/c/Windows/System32/OpenSSH/:/mnt/c/MinGW/bin:/mnt/c/Program Files/nodejs/:/mnt/c/Program Files/Git/cmd:/mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin/:/mnt/c/Users/Sai Pavani/AppData/Local/Programs/Python/Launcher/:/mnt/c/Users/Sai Pavani/AppData/Local/Microsoft/WindowsApps/:/mnt/c/Users/Sai Pavani/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/Sai Pavani/AppData/Roaming/npm:/mnt/c/ProgramData/Sai Pavani/GitHubDesktop/bin:/mnt/c/Users/Sai Pavani/AppData/Local/GitHubDesktop/bin:/mnt/c/Users/Sai Pavani/AppData/Local/Python/bin:/snap/bin

Parent waiting for child...
Child Process Started
Child PID: 2001
shellforge          skill14_append.txt    skill4    skill7       skill_12      skill_15      skill_18      skill_21      skill_24      temp
shellforge.c        skill14_error.txt     skill4.c  skill7.c     skill_12.c.c  skill_15.c.c  skill_18.c.c  skill_21.c.c  skill_24.c.c
simplecommandloop   skill15_combined.txt  skill5    skill_10     skill_13      skill_16      skill_19      skill_22      skill_8
simplecommandloop.c skill21_errors.log    skill5.c  skill_10.c.c skill_13.c.c  skill_16.c.c  skill_19.c.c  skill_22.c.c  skill_8.c.c
skill13_input.txt   skill3                skill6    skill_11     skill_14      skill_17      skill_20      skill_23      skill_9
skill13_output.txt  skill3.c              skill6.c  skill_11.c.c skill_14.c.c  skill_17.c.c  skill_20.c.c  skill_23.c.c  skill_9.c.c
Child exited with status: 0
sai_pavani@Sai-Pavani:~/OS_SKILL$
```